

UNIT 1 – ELECTRIC CHARGE, FORCE, AND FIELD

1.1 Coulomb's Law and Superposition**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

1. Two point charges, $q_1 = +3.0 \mu\text{C}$ and $q_2 = -5.0 \mu\text{C}$, are 0.20 m apart. Find the magnitude of the force each exerts on the other and state whether it is attractive or repulsive.

$$F = \frac{k|q_1q_2|}{r^2} = \frac{(8.99 \times 10^9)(3.0 \times 10^{-6})(5.0 \times 10^{-6})}{(0.20)^2} = 3.4 \text{ N. Opposite signs, so the force is attractive.}$$

2. Three charges lie on the x -axis: $+2.0 \mu\text{C}$ at $x = 0$, $-2.0 \mu\text{C}$ at $x = 0.10 \text{ m}$, and $+4.0 \mu\text{C}$ at $x = 0.30 \text{ m}$. Find the net force (magnitude and direction) on the $+4.0 \mu\text{C}$ charge.

$$\begin{aligned} \text{From the } +2.0 \mu\text{C} \text{ charge (repulsive, } +x\text{): } F_1 &= \frac{k(2.0 \times 10^{-6})(4.0 \times 10^{-6})}{(0.30)^2} = 0.80 \text{ N. From} \\ \text{the } -2.0 \mu\text{C} \text{ charge (attractive, } -x\text{): } F_2 &= \frac{k(2.0 \times 10^{-6})(4.0 \times 10^{-6})}{(0.20)^2} = 1.80 \text{ N. Net: } F = \\ 0.80 - 1.80 &= -1.0 \text{ N, i.e. } \mathbf{1.0 \text{ N toward } -x}. \end{aligned}$$

3. Two identical small spheres carry charges q and $3q$ and repel with force F . They are touched together and separated to the same distance. Find the new force in terms of F .

After contact each carries $2q$. The product of charges goes from $3q^2$ to $4q^2$, so the new force is $\frac{4}{3}F$.